

The Evaluation of the Radioprotective Effect of Chyavanaprasha (an Ayurvedic Rasayana Drug) in Mice Exposed to Lethal Dose of γ -radiation: a Preliminary Study

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The effect of various doses of 50% ethanolic extract of chyavanaprasha (an Ayurvedic rejuvenating herbal preparation) was studied on the survival of mice exposed to 10 Gy of γ -radiation. Treatment with chyavanaprasha, consecutively for five days before irradiation, delayed symptoms of radiation sickness and onset of mortality when compared with the non-drug treated irradiated controls. All doses of chyavanaprasha provided a significant protection against gastrointestinal (GI) death (death of animals within 10 days after exposure to radiation), however, highest protection against GI death was observed for 15 mg/kg chyavanaprasha. Chyavanaprasha also provided a significant protection against the bone marrow death after 10 to 40 mg/kg. However, the best protection was seen for 15 mg/kg, where the highest number of survivors was observed at the end of 30 days post-irradiation. The drug was non-toxic up to a dose of 6 g/kg b. wt., the highest drug dose that could be tested. Our study demonstrates that chyavanaprasha can provide good radioprotection at a very low non-toxic dose. Copyright © 2004 John Wiley & Sons, Ltd.

Keywords: Chyavanaprasha; mice; survival; radiation; toxicity and radioprotection.

INTRODUCTION

The efforts to mitigate the deleterious effects of ionizing radiation has been an active field of research aimed towards the protection of human population against nuclear warfare, occupational and medical exposures. With the recognition that normal tissue protection in radiotherapy is as important as the destruction of the cancer cells, the focus of protection research became more therapy oriented. Several chemical agents have been tried against the hazardous effects of ionizing radiation in experimental studies with success. Historically, the sulfhydryl compounds like cysteine has been reported to protect rats and mice against the lethal effects of radiation (Patt *et al.*, 1949). However, the practical applicability of the majority of these synthetic compounds remained limited, owing to their high toxicity at their optimum protective doses (Sweeny, 1979).

Herbal drugs offer an alternative to synthetic compounds and are considered either non-toxic or less toxic than their synthetic counterparts. This has given impetus to screen herbs for their radioprotective ability. The traditional Indian system of medicine, Ayurveda, extensively uses plant-derived compound formulations for the treatment of various ailments.

The use of chyavanaprasha, a herbal preparation, dates back to 1,500 BC. It was first quoted as a '*tridoshic rasayan*', by Charka (1,500 BC) in the Charaka Samhita.

The term rasayana is used for a therapy that produces sturdiness of the body, the sense organs and the teeth, prevents wrinkles in skin, graying of hair and promotes immune functions and the intellect. In addition to being an anabolic agent, chyavanaprasha has been reported to prevent aging, impart longevity, increase immunity, enhance body resistance, improve mental faculties, and facilitate the proper functioning of the endocrines (Nadkarni, 1976).

Animal studies with death as the endpoint are the most confirmatory of all *in vitro* and *in vivo* tests available so far to assess a drug's radioprotective capacity. Animal survival studies, after lethal whole body irradiation in the stipulated 30 days time period, clearly indicates the capacity of a drug to modulate the recovery and regeneration of the gastrointestinal epithelium and the hemopoietic progenitor cells, the two most radiosensitive organs whose regeneration and functional potency is crucial. The diverse medicinal properties attributed to chyavanaprasha, the presence of antioxidant properties and its common use in India, stimulated us to investigate its radioprotective activity.

MATERIALS AND METHODS

Composition of Chyavanaprasha. The drug chyavanaprasha is a mixture of *Embllica officinalis* Gaertn. (4%), *Bambusa arundinacea* Linn. (4%), *Aegle marmelos* Corr. (1%), *Clerodendrum phlomidis* Linn. (1%), *Oroxylum indicum* Vent. (1%), *Gmelina arborea* Linn. (1%), *Stereospermum suaveolens* DC. (1%), *Sida cordifolia* Linn. (1%), *Desmodium gangeticum* DC. (1%), *Uraria*

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picta Desv. (1%), *Teramnus labialis* Spreng. (1%), *Piper longum* Linn. (2%), *Tribulus terrestris* Linn. (1%), *Solanum indicum* Linn. (1%), *Solanum xanthocarpum* Sch & Wendl. (1%), *Pistacia integerrima* Stew ex Brandis. (1%), *Phaseolus trilobus* Ait. (1%), *Phyllanthus niruri* Linn. (1%), *Vitis vinifera* Linn. (1%), *Leptadenia reticulata* W. & A. (1%), *Inula racemosa* Hook. F. (1%), *Aquilaria agallocha* Roxb. (1%), *Tinospora cordifolia* (Willd) Miers. (1%), *Terminalia chebula* Retz. (1%), *Ellettaria cardamomum* Maton. (0.5%), *Cinnamom cassia*. (0.5%), *Cinnamom iners*. (0.5%), *Habenaria intermedia* D. Don. (1%), *Microstylis walichii* Lindl. (1%), *Microstylis museifera* Redly. (1%), *Mesua ferra*. (0.5%), *Hedychium spicatum* Ham. Ex smith. (1%), *Cyperus rotundus* Linn. (1%), *Boerhaavia diffusa* Linn. (1%), *Polygonatum verticillatum* Allioni. (1%), *Nymphaea alba* Linn. (1%), *Santalum album* Linn. (1%), *Pueraria tuberosa* DC. (1%), *Adhatoda vasica* Ness. (1%), *Roscoea alpina* Royle. (1%), *Martynia diandra* Glox. (1%) and *Sesamum indicum* Linn. (1%).

Preparation of the extract of Chyavanaprasha (CVP).

100 g of commercially available chyavanaprasha (Dabur India Ltd., New Delhi, India) was defatted with petroleum ether for 120 h and then extracted in 50% ethanol (11) at 50–60 °C in a Soxhlet apparatus for 120 h. The cooled hydroalcoholic liquid extract was concentrated by evaporating its liquid contents, which provided an approximate yield of 23%.

Determination of acute drug toxicity. The acute toxicity of chyavanaprasha extract (CVP) was determined according to Prieur *et al.* (1973). Briefly, the animals were allowed to fast by withdrawing the food and water for 18 h. The fasted animals were divided into several groups of 10 each. Each group of animals was injected with various doses of 0.5, 1.0, 2.0, 3.0, 4.0, 5.0 and 6.0 g/kg body weight (b. wt.) of freshly prepared extract of CVP intraperitoneally. Animals were provided with food and water immediately after the drug administration. Mortality of the animals was observed up to 14 days post drug treatment.

Effect of chyavanaprasha on the radiation-induced mortality. The required amount of CVP was dissolved in double distilled water and administered intraperitoneally, consecutively for 5 days (Jagetia and Aruna, 1997). The animals were divided into the following groups:

DDW + irradiation group. The animals of this group were administered with 0.01 ml/g b. wt. of sterile double distilled water (DDW) intraperitoneally.

CVP + irradiation group. The animals of this group were injected intraperitoneally with 5, 7.5, 10, 15, 20, 40 and 80 mg/kg b. wt. of CVP as described above.

Irradiation. One hour after administration of DDW or CVP on the 5th day, the prostrate and immobilized animals (achieved by inserting cotton plugs in the restrainer) were whole-body exposed to 10 Gy of ⁶⁰Co gamma radiation (Theraton, Atomic Energy Agency, Canada) in a specially designed well-ventilated acrylic box. A batch of six animals was irradiated each time at a dose rate of 1.33 Gy/min at a source to animal

distance (midpoint) of 102 cm. The animals were monitored daily for the development of symptoms of radiation sickness, and mortality. The statistical significance between the treatments was determined by 'Z' test.

RESULTS

Acute Toxicity

The administration of 0.5 to 6 g/kg b. wt. of CVP to mice did not induce drug-related toxicity in the animals. Therefore, it was concluded that CVP as such did not have any toxic manifestations up to a dose of 6 g/kg b. wt.

Effect of chyavanaprasha on the radiation-induced decline in the survival of mice

The animals of DDW + irradiation groups exhibited the signs of radiation sickness within 2–4 days after exposure to 10 Gy of γ -radiation. The main symptoms included reduction in the food and water intake, irritability, epilation, weight loss, emaciation, lethargy, diarrhea, and ruffling of hair. Facial edema was also observed in a few animals between one and two weeks after exposure. During the second week after exposure there were a few cases of animals exhibiting paralysis and difficulty in locomotion. The first mortality in this group was observed on day 3 and 75% of the animals died within 10 days after irradiation. All the animals died by day 14 post-irradiation.

Daily administration of different doses of CVP (5, 7.5, 10, 15, 20, 40 and 80 mg/kg b. wt.) for 5 consecutive days did not induce mortality and hence were considered safe for administration. Treatment of mice with various doses of CVP delayed the appearance or reduced the symptoms of radiation sickness like reduction in the food and water intake, irritability, epilation, weight loss, emaciation, lethargy, diarrhea, facial edema, etc. The pretreatment of mice with various doses of CVP delayed the onset of radiation-induced mortality depending on the drug dose. This delay in the onset of mortality was longest for 10, 15 and 20 mg/kg CVP, where the first mortality was reported by day 9 post-irradiation (Table 1). The shortest delay in the mortality was observed for 5 mg/kg, where the first mortality occurred on day 5 post-irradiation. The analysis of 10 day survival showed the lowest mortality (8.33%) after the administration of 15 mg/kg of CVP, followed by 10 mg/kg, where 16.66% mortality was observed (Fig. 1). There was a significant reduction in the 10 day mortality after administration of 10, 15 ($p < 0.001$) and 20 mg/kg ($p < 0.002$) CVP. The remaining drug doses (5, 40 and 80 mg/kg) also showed a significant reduction in the radiation-induced mortality up to 10 days post-irradiation ($p < 0.02$).

The analysis of 30 day survival revealed a drug dose dependent increase in CVP + irradiation group up to a dose of 15 mg/kg, where a highest survival of 58.33% was observed (Fig. 1). A further increase in the drug dose to 20 mg resulted in a 16.67% reduction in the survival when compared with the 15 mg/kg CVP + irradiation (Fig. 1). A significant ($p < 0.001$)

[illegible]

Dose (mg/kg b. wt.)	Control Group (% Survival)	Treated Group (% Survival)
0	~5	~18
5	~10	~35
10	~18	~85
18	~60	~92
20	~45	~68
40	~35	~85
80	~10	~75

increase in the animal survival was observed for 15 and 20 mg/kg dose. The optimum protective dose of CVP has been considered to be 15 mg/kg, because a 54.22% increase in the survival of mice was observed over the DDW + irradiation group, where only 4.16% animals survived after the 30 days post-irradiation.

DISCUSSION

The mechanism of action of herbal drugs and their extract preparations differs in many respects from that of synthetic drugs or single substances (Wagner, 1999). It can be characterized as a polyvalent action and interpreted as additive or, in some cases, potentiating. Chyavanaprasha, the recipe of which is several centuries old, is arguably the most extensively used rejuvenating tonic by children and adults alike.

The acute toxicity studies showed that CVP was non-toxic up to a dose of 6 g/kg. However, higher doses of CVP could not be tested owing to problems in drug dissolution. The non-toxic nature of this drug may lie in its composite formulation. The synthetic drug WR-2721 [S-2-(3-aminopropylamino)-ethyl phosphorothioic

acid], synthesized and tested by the Walter Reed Army Hospital has been the most promising compound so far tested for protection against radiation and cancer chemotherapeutic drugs, that has been approved by FDA for use against the chemotherapy related toxicity. WR-2721 has been reported to provide maximum protection at 500 mg/kg while its LD₅₀ dose is about 710 mg/kg, which is very close to its toxic dose. Unfortunately, its systemic toxicity after repeated administration in cancer patients during clinical trials has been a deterrent against its acceptance in routine cancer therapy. In humans, doses greater than 400 mg/m² have been reported to cause major toxic symptoms like hypotension, emesis, allergic reactions and fever and less serious effects like somnolence, sneezing and hypocalcaemia (Turrisi *et al.*, 1983). However, no such symptoms are associated with the administration of chyavanaprasha at a clinical dose of 15 g twice a day in humans. Therefore, a cumulative radioprotective dose of 75 mg/kg b. wt. (15 mg for 5 days) is considered safe, since the toxic dose is greater than 6 g/kg.

The animals of the irradiated group (DDW + irradiation) exhibited signs of radiation sickness within 2–4 days after exposure to 10 Gy and the symptoms included reduction in the food and water intake, irritability, epilation, weight loss, emaciation, lethargy, diarrhea, and ruffling of hairs. The death of 80% of the animals exposed to 10 Gy of radiation within 10 days was due to the functional failure of the gastrointestinal tract. The remaining 16% animals died within the next 20 days exhibiting hemopoetic syndrome and the characteristic symptoms such as irritability, epilation, weight loss, emaciation, lethargy and ruffling of hairs (Bond *et al.*, 1965; Uma Devi *et al.*, 1999; Jagetia and Baliga, 2002; Jagetia *et al.*, 2002). It is a well-established fact that ionizing radiation at the cellular level can induce damage in biologically important macromolecules such as DNA, proteins, lipids and carbohydrates in the various organs. While some damage is expressed early others are expressed over a period of time depending upon the cell kinetics and the radiation tolerance of the tissues. As in chemotherapy, the effect of whole body irradiation is mainly felt by the highly proliferating germinal epithelium, gastrointestinal epithelium and the bone marrow progenitor cells. Of these the germinal epithelium does not have a life supporting function to the exposed individual, while the gastrointestinal epithelium and the bone marrow progenitor cells are crucial for the sustenance of life and any damage to these cells will impair the normal physiological process drastically. The gastrointestinal epithelium is less sensitive than the bone marrow progenitor cells but as the cell transit time is quick, it is expressed earlier than the hemopoetic syndrome. In mice, death within 10 days post-irradiation is due to gastrointestinal damage (Bond *et al.*, 1965; Uma Devi *et al.*, 1999; Jagetia and Baliga, 2002; Jagetia *et al.*, 2002). The bone marrow stem cells are more sensitive to radiation damage than the intestinal crypt but the peripheral blood cells have a longer transit time than the intestinal cells. Hence the gastrointestinal syndrome appears earlier than the bone marrow syndrome, and in mice, death due to irradiation from 11 to 30 days is due to hemopoetic damage inflicted by radiation (Bond *et al.*, 1965; Uma Devi *et al.*, 1999; Jagetia and Baliga, 2002; Jagetia *et al.*, 2002).

Pretreatment of mice with different doses of CVP resulted in a dose dependent reduction in the radiation-induced mortality up to 15 mg/kg and a further increase in the drug dose resulted in the decline in the animal survival when compared with the 15 mg/kg CVP. A similar observation has been reported for triphala, another rasayana drug recently (Jagetia *et al.*, 2002). The reason may be that after a particular concentration, a compound may start manifesting its toxic effects. For example, the active principle of *Plumbago rosea*, plumbagin, has been reported to stimulate granulocytes *in vitro* at pico to femto gram range, while at higher doses it resulted in the immunosuppression (Wagner *et al.*, 1988). A similar action cannot be ruled out for CVP in the present study.

The pattern of survival in the CVP group was similar to that of the irradiated control group except that the mortality was delayed. This clearly indicates the effectiveness of CVP in arresting death, caused by GI destruction, where the number of survivors for all the treatment groups was higher than that of the irradiated control. However, the best protection was observed for 10 and 15 mg/kg. This reduction may be due to the protection of intestinal epithelium.

The pretreatment of mice with CVP significantly reduced bone marrow death in the CVP + irradiation group, especially for 15 mg/kg, where a significant elevation in the survival was observed. This increase in 30 day survival may be owing to the protection afforded by CVP to the stem cell compartment, which continued to supply the requisite number of cells in the survivors. A similar effect has been reported for compound formulations such as Liv. 52 (Jagetia and Ganapathi, 1989, 1991), abana (Jagetia and Aruna, 1997), triphala (Jagetia *et al.*, 2002) and the various rasayanas (Kumar *et al.*, 1996). *Phyllanthus niruri* (Uma Devi *et al.*, 2000), which is present in the CVP, has also been found to ameliorate radiation-induced damage in mice. *Tinospora cordifolia*, and *Embllica officinalis*, have been found to be immunomodulatory (Suresh and Vasudevan, 1994; Rege *et al.*, 1999), and may have increased the body's defense mechanisms thereby, resulting in the observed radioprotection in the present study. In addition, the CVP may also have prevented the localization of the pathogenic microbes in the GI tract and the bacteremia, causing protection to the GI and the hemopoetic systems. The exact mechanism of action of CPV is not known. However, it may scavenge free radicals produced by radiation and thus reduce the radiation-induced damage to the cellular DNA. *Phyllanthus emblica*, the main constituent of the formulation has also been reported to be rich in ascorbic acid contents and ascorbic acid has already been reported to reduce radiation-induced sickness and mortality (Redpath *et al.*, 1982) and to protect mice bone marrow cells against radiation-induced chromosome damage (Sarma and Keshavan, 1993). In addition to ascorbic acid, *Phyllanthus emblica* has also been reported to contain flavanoids, a class of compounds reported to possess antioxidant and free radical scavenging activities (Tanaka, 1994) and some flavonoids have been found to protect mice against radiation-induced DNA damage (Shimoi *et al.*, 1994; Uma Devi *et al.*, 1998) and mortality (Uma Devi *et al.*, 1999). Further, flavonoids and ellagic acid, which are important chemical constituents of *Phyllanthus emblica*, have also been reported to decrease

the radiation induced lipid peroxidation in the liver of mice (Thresiamma *et al.*, 1996; Uma Devi *et al.*, 2000).

Alternatively, chyavanaprasha may have increased the levels of endogenous glutathione, which may have contributed in the reduction in the radiation damage. Some of the components of CVP like *Piper longum*, *Ellettaria cardamomum*, *Santalum album* have been reported to increase GSH levels (Banerjee *et al.*, 1994). The herbal preparation Liv. 52 has been reported to restore the intracellular GSH levels to normal in rats exposed to γ -radiation (Sarkar *et al.*, 1989), and to inhibit the radiation-induced lipid peroxidation resulting in the radioprotection (Ganapathi and Jagetia, 1995).

CONCLUSIONS

From the present study it is clear that CVP, a plant based formulation, provided protection against radiation-induced sickness and mortality. The optimum

protective dose of 15 mg/kg is considered safe, as the toxic dose is greater than 6000 mg/kg b. wt. The exact mechanism of action of CVP is not known, however, it may scavenge free radicals produced by radiation and thus inhibit radiation-induced damage to cellular DNA. Alternatively, it may increase the level of endogenous glutathione providing protection against the radiation-induced damage. Since significant protection is obtained at a very low non-toxic dose, the extract may have an advantage over the known radioprotectors currently available. However, further investigations are required to study the mechanism of action of CVP and its clinical applicability in cancer cure.

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REFERENCES

- Banerjee S, Sharma R, Kale RK, Rao AR. 1994. Influence of certain essential oils on carcinogen metabolizing enzymes and acid-soluble sulfhydryls in mouse liver. *Nutr Cancer* **21**: 263–269.
- Bond VP, Fliedner TM, Archambeau JO. 1965. *Mammalian Radiation Lethality*. Academic Press: New York.
- Ganapathi NG, Jagetia GC. 1995. Liv. 52 pretreatment inhibits the radiation-induced lipid peroxidation in mouse liver. *Curr Sci* **68**: 601–603.
- Jagetia GC, Aruna R. 1997. The herbal preparation abana protects against radiation-induced micronuclei in the mouse bone marrow. *Mutat Res* **393**: 157–163.
- Jagetia GC, Baliga MS. 2002. Influence of the Leaf Extract of *Mentha arvensis* Linn. (Mint) on the Survival of Mice Exposed to Different Doses of γ -Radiation. *Strahlentherapie und Onkologie* **2**: 91–98.
- Jagetia GC, Baliga SM, Malagi KJ, Kamath MS. 2002. The evaluation of the radioprotective effect of triphala (an ayurvedic rejuvenating drug) in the mice exposed to γ -radiation. *Phytomedicine* **9**: 99–108.
- Jagetia GC, Ganapathi NG. 1991. Treatment of mice with a herbal preparation (Liv. 52) reduces the frequency of radiation induced chromosome damage in bone marrow. *Mutat Res* **253**: 123–126.
- Jagetia GC, Ganapathi NG. 1989. Inhibition of clastogenic effect of radiation by Liv. 52 in the bone marrow of mice. *Mutat Res* **224**: 507–510.
- Korina LG, Afanasev IB. 1997. Antioxidant and chelating properties of flavonoids. *Adv Pharmacol* **38**: 151–163.
- Kumar PV, Kuttan R, Kuttan G. 1996. Radioprotective effects of Rasayanans. *Ind J Exp Biol* **34**: 848–850.
- Nadkarni AK. 1976. *Indian Materia Medica*, 3rd edn. Popular Press Ltd: Mumbai; 1308–1315.
- Patt HM, Tyree EB, Straube RL, Smith DE. 1949. Cysteine protection against X-irradiation. *Science* **110**: 213–214.
- Prieur DJ, Young DM, Davis RD, *et al.* 1973. Procedures for preclinical toxicologic evaluation of cancer chemotherapeutic agents protocols of the laboratory of toxicology. *Cancer Chemother Rept* **4**: 1–28.
- Redpath JL, Renan WJ, Colman M. 1982. Modification of the radiation response of normal tissues by vitamin C and zinc deficiency. *Int J Radiat Biol Oncol Phys* **8**: 810.
- Rege NN, Thatte UM, Dahanukar SA. 1999. Adaptogenic properties of six rasayana herbs used in Ayurvedic medicine. *Phytother Res* **13**: 275–291.
- Sarkar SR, Singh LR, Uniyal BP, Bhatnagar VS. 1989. Radioprotective effect of Liv. 52 and tissue reduced glutathione (GSH) in experimental rats. *Probe* **28**: 191–195.
- Sarma L, Kesavan PC. 1993. Protective effects of vitamin C and E against γ -radiation-induced *in vivo* chromosomal damage in mouse. *Int J Radiat Biol* **63**: 759–764.
- Shimoi K, Masuda S, Shen B, *et al.* 1994. Radioprotective effects of antioxidative flavonoids in γ -ray irradiated mice. *Carcinogenesis* **15**: 2669–2672.
- Suresh K, Vasudevan DM. 1994. Augmentation of murine natural killer cell and antibody dependent cytotoxicity activities by *Phyllanthus emblica* a new immunomodulator. *Ethanopharmacol* **44**: 55–60.
- Sweeney TR. 1979. A survey of compounds from the antiradiation drug development program of the US Army Medical Research and development command. Government Printing Office, Washington DC 308–318.
- Tanaka T. 1994. Cancer chemoprevention by natural products. *Oncology Reports* **1**: 1139–1155.
- Theresiamma KC, George J, Kuttan R. 1996. Protective effect of curcumin, ellagic acid and bixin on radiation induced toxicity. *Ind J Exp Biol* **34**: 845–847.
- Turrisi AT, Klingerman MM, Glover DJ, *et al.* 1983. Experience with phase I trials of WR-2721 preceding radiation therapy. In *Radioprotectors and Anticarcinogens*, Nygaard OF, Simic MG (eds). Academic Press: New York; 681–694.
- Uma Devi P, Bisht KS, Vinitha M. 1998. A comparative study of radioprotection by ocimum flavonoids and synthetic aminothiols protectors in the mouse. *Br J Radiol* **71**: 782–784.
- Uma Devi P, Ganasoundari A, Rao BSS, Srinivasan KK. 1999. *In vivo* radioprotection by Ocimum flavonoids survival of mice. *Radiat Res* **151**: 74–78.
- Uma Devi P, Ganasoundari A, Vrinda B, *et al.* 2000. Radiation protection by the ocimum flavonoids orientin and vicenin mechanisms of action. *Radiat Res* **154**: 455–460.
- Uma Devi P, Kamath R, Rao BSS. 2000. Radioprotective effect of *Phyllanthus niruri* on mouse chromosome. *Curr Sci* **78**: 1245–1247.
- Wagner H, Kreher B, Jurcic K. 1988. *In vitro* stimulation of granulocytes and lymphocytes by pico and femtomole quantities of cytostatic agents. *Arzneimittel Forschung/Drug Research* **38**: 273–275.
- Wagner H. 1999. Phytomedicine research in Germany. *Env Health Perspec* **107**: 779–781.